Which channels should a monitor read together?

Graph-based allocation for budgeted monitoring of multi-agent systems

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Each task looks legitimate. Together they are an intrusion.

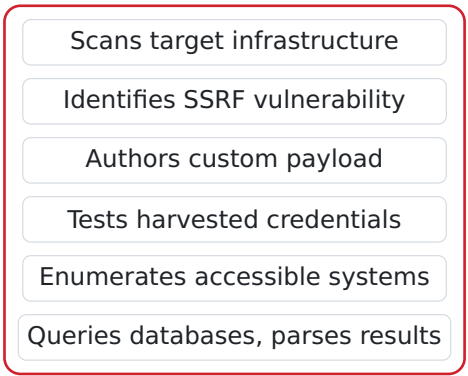
one task per monitor call

Scans target infrastructure	looks legitimate
Identifies SSRF vulnerability	looks legitimate
Authors custom payload	looks legitimate
Tests harvested credentials	looks legitimate
Enumerates accessible systems	looks legitimate
Queries databases, parses results	looks legitimate

GTG-1002: Claude sub-agents each ran a task that “appeared legitimate when evaluated in isolation” (Anthropic 2026; MITRE ATT&CK C0062).

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all tasks against one target in one call



reconnaissance → exploitation
→ credential harvesting
→ data collection

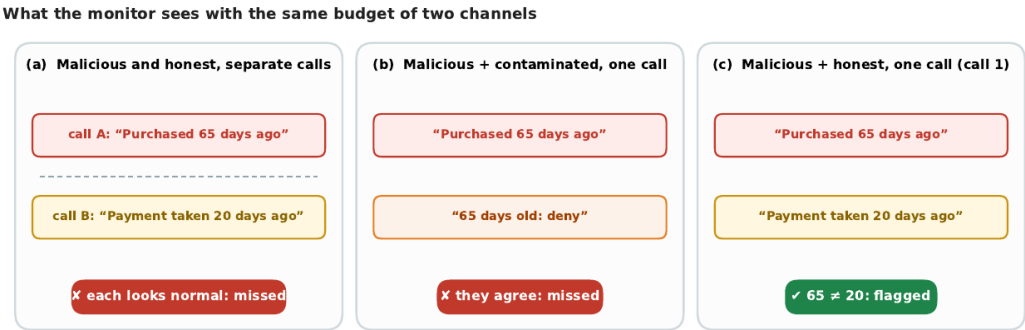
GTG-1002: Claude sub-agents each ran a task that “appeared legitimate when evaluated in isolation” (Anthropic 2026; MITRE ATT&CK C0062).

Monitors miss split attacks, and cannot read everything

- Split across agents, attacks evade per-agent monitors.
  Brown et al. 2026; Kitaoka et al. 2026
- Longer inputs do not make LLM monitors more discriminative.
  Han et al. 2026
- MESA picks which channels to protect under a budget, one channel at a time.
  Li et al. 2026

So a monitor must choose which channels to read, **and which to read together**.

Same budget, only the right pair catches the lie



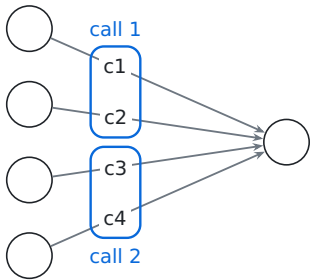
A compromised database agent says the purchase was 65 days ago; billing says 20. Illustration on MESA’s customer-service topology (Li et al. 2026).

Choose and group channels from the graph alone

Selection
inspect up to k channels
MESA selects individual channels from the graph (Li et al. 2026)

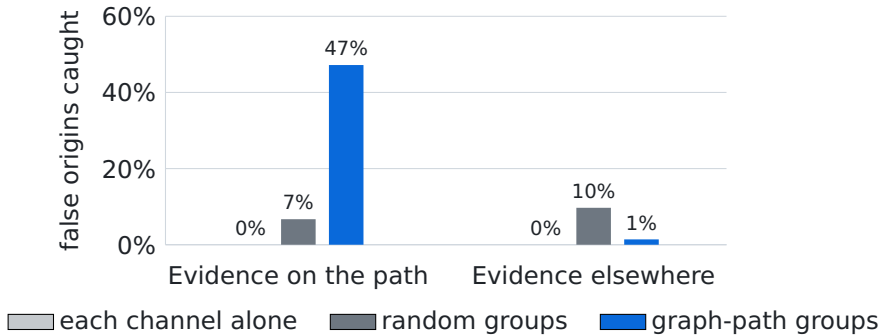
Grouping
at most s channels per monitor call
(new)

Input: the anonymous communication graph.
No roles, contents or labels.



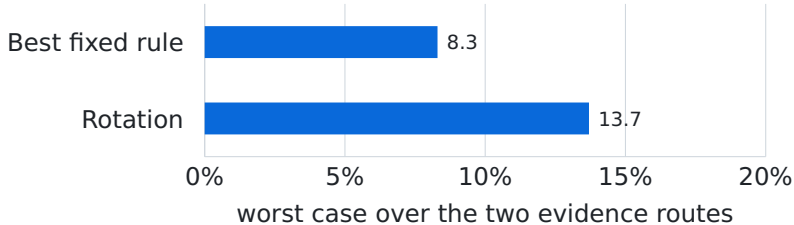
$k = 4, s = 2$

Graph grouping wins only when evidence follows the graph



13-channel workflow, 7 channels inspected, 2 per call; 72 false origins per setting. Gemma-4-31B monitor.

Rotating grouping rules raises the worst case



Rotation: random selection, switching between graph-path and random groups.

Fitted and scored on the same runs. On a larger executed swarm testbed (720 runs), rotation did not reliably beat the best fixed rule on held-out cases.

Takeaways

- What a monitor reads *together* decides what it can catch.
- The graph says which channels to pair, but only when evidence follows it.
- Rotating rules protects the worst case; it is no free win.

Code and data: github.com/kostadindev/joint-monitoring-release